

NHỮNG ỨNG DỤNG ĐƠN GIẢN CỦA ĐẠI SỐ TUYẾN TÍNH

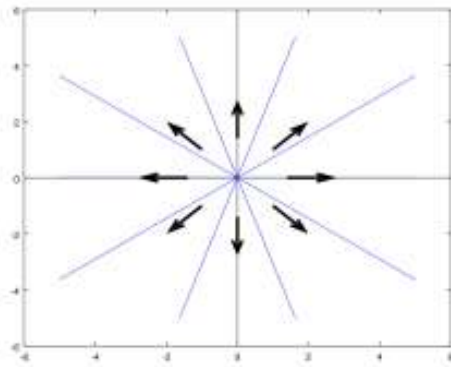
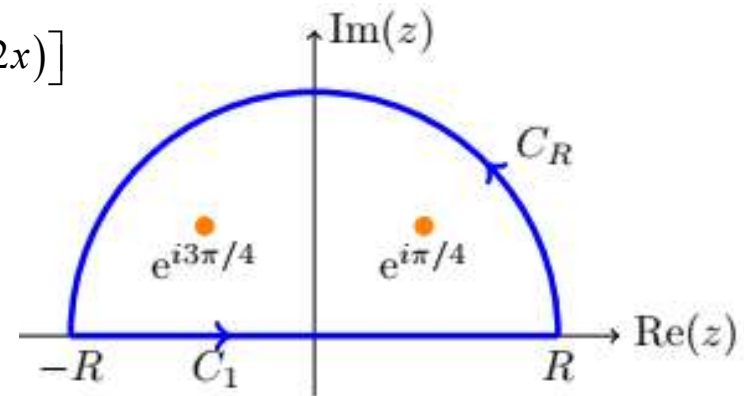
I. SỐ PHỨC

$$1) I = \int e^x \cos(2x) dx ? \rightarrow I = \frac{1}{5} e^x [\cos(2x) + 2 \sin(2x)]$$

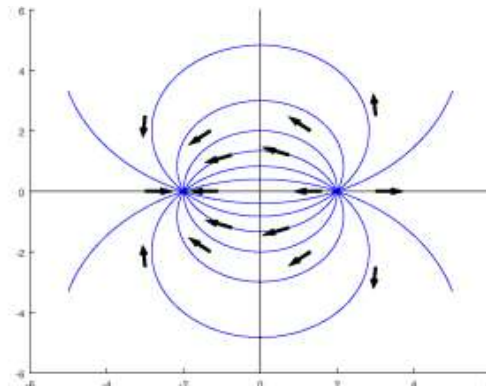
$$2) I = \int \frac{x+1}{x^2+1} dx ? \rightarrow I = \frac{1-i}{2(x-i)^2} + \frac{1+i}{2(x+i)^2}$$

$$3) I = \int_{-\infty}^{\infty} \frac{1}{x^4+1} dx \rightarrow I = \frac{\sqrt{2}}{2} \pi$$

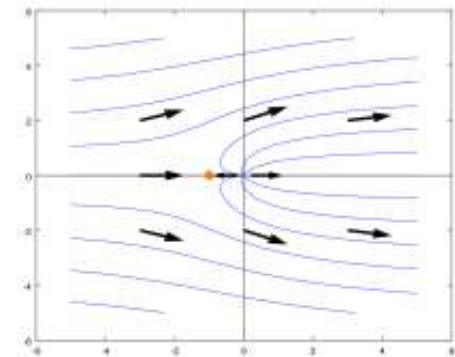
4) Lapce and Fourier transform



$$\Phi(z) = \log(z)$$



$$\Phi(z) = \log(z-2) - \log(z+2)$$



$$\Phi(z) = z + \frac{Q}{2\pi} \log(z)$$

II. MA TRẬN

$$AX = B \rightarrow X = A^{-1}B$$

$$z = a + bi = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \rightarrow z_1 z_2$$

1) Nhân hai vector

$$(1 \quad 2 \quad 5) \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} = 30$$

$$Gia: \begin{cases} I: 1000 \\ II: 1500 \\ III: 2000 \end{cases} \quad So.luong: \begin{cases} I: 50 \\ II: 80 \\ III: 12 \end{cases} \Rightarrow (1000 \quad 1500 \quad 2000) \begin{pmatrix} 50 \\ 80 \\ 12 \end{pmatrix} = ?$$

$$Diem: \begin{cases} T: 3 \\ H: 1 \\ Th: 0 \end{cases} \quad Tran: \begin{cases} T: 32 \\ H: 2 \\ Th: 4 \end{cases} \Rightarrow Tong.Diem: (3 \quad 1 \quad 0) \begin{pmatrix} 32 \\ 2 \\ 4 \end{pmatrix} = ?$$

2) Nhân vector & ma trận

$$(1 \quad 2 \quad 5) \begin{pmatrix} 10 & 0 & 11 \\ 20 & 40 & 6 \\ 8 & 15 & 25 \end{pmatrix} = (? \quad ? \quad ?)$$

cot ~ ngay

$$(3 \quad 1 \quad 0) \begin{pmatrix} 33 & 31 & 5 \\ 2 & 1 & 6 \\ 3 & 6 & 27 \end{pmatrix} = (? \quad ? \quad ?)$$

cot ~ doi.bong

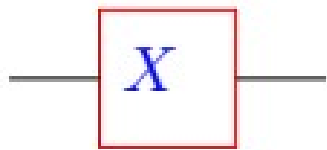
3) Nhân hai ma trận

$$\begin{pmatrix} * & * & * \\ * & * & * \\ * & * & * \end{pmatrix} \begin{pmatrix} \# & \# & \# \\ \# & \# & \# \\ \# & \# & \# \end{pmatrix} = ?$$

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \begin{bmatrix} 1/E_1 & -\nu_{21}/E_2 & 0 \\ & 1/E_2 & 0 \\ & & 1/G_{12} \end{bmatrix} \cdot \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} \quad \begin{aligned} \epsilon_{ij} &= \epsilon_{ij}^M + \epsilon_{ij}^\theta \\ \sigma_{ij} &= C_{ijkl}\epsilon_{kl}^M = C_{ijkl}(\epsilon_{kl} - \epsilon_{kl}^\theta) \end{aligned}$$

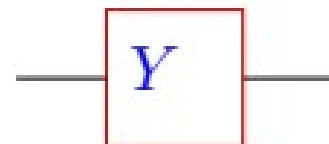
$$F_{\mu\nu} = \eta_{\alpha\nu}F^{\beta\alpha}\eta_{\mu\beta} = \begin{bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{bmatrix}$$

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow S_z \begin{pmatrix} a \\ b \end{pmatrix} = \lambda \begin{pmatrix} a \\ b \end{pmatrix} \rightarrow \det \begin{pmatrix} \frac{\hbar}{2} - \lambda & 0 \\ 0 & -\frac{\hbar}{2} - \lambda \end{pmatrix} = 0 \rightarrow \lambda = \pm \frac{\hbar}{2} \rightarrow \begin{cases} \chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases}$$



$$X|0\rangle = |1\rangle \quad X|1\rangle = |0\rangle$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



$$Y|0\rangle = i|1\rangle \quad Y|1\rangle = -i|0\rangle$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

a) *Phép quay 2 chiều* $R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ $* \vec{r}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \theta = 30^\circ, 120^\circ$ $** \vec{r}_2 = \begin{pmatrix} 4 \\ -1 \\ 6 \end{pmatrix}, \theta = 45^\circ, 60^\circ$

b) *Phép quay 3 chiều* $R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$ $R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$ $R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

c) *Phép tịnh tiến* $(x, y, 1) \rightarrow (x+a, y+b, 1)$.

$$\begin{pmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x+a \\ y+b \\ 1 \end{pmatrix}$$

Application of matrix in computer graphics

d) *Phóng to, thu nhỏ* $(x, y, 1) \rightarrow (sx, ty, 1)$.

$$\begin{pmatrix} s & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} sx \\ ty \\ 1 \end{pmatrix}$$

2.6: Sử dụng ma trận để mã hóa tin nhắn

1)

A	B	C	D	E	F	G	H	I	J	K	L	M	
1	2	3	4	5	6	7	8	9	10	11	12	13	Space
N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
14	15	16	17	18	19	20	21	22	23	24	25	26	27

Sử dụng ma trận $A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$

để mã hóa tin nhắn ATTACK NOW

$$\begin{bmatrix} 41 \\ 61 \end{bmatrix} \begin{bmatrix} 22 \\ 23 \end{bmatrix} \begin{bmatrix} 25 \\ 36 \end{bmatrix} \begin{bmatrix} 55 \\ 69 \end{bmatrix} \begin{bmatrix} 61 \\ 84 \end{bmatrix}$$

2)

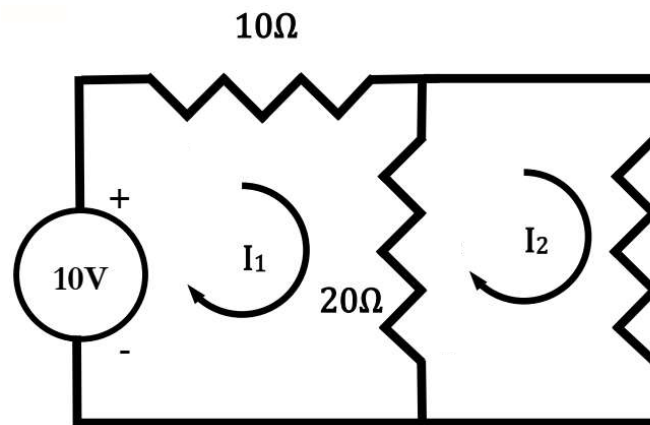
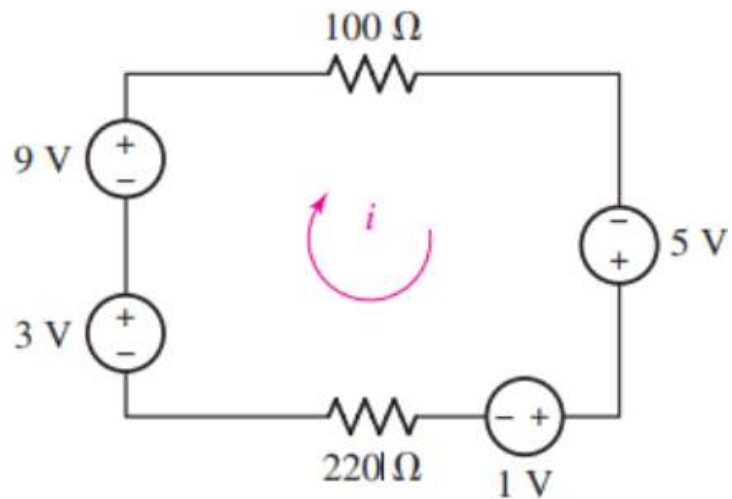
space	A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z	
14	15	16	17	18	19	20	21	22	23	24	25	26	

a) Sử dụng 2 vectors X và Y để diễn tả tin nhắn ACTION

b) Sử dụng ma trận A để mã hóa tin nhắn này

251,223,190,284,204,216,187,234

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \& Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} \quad A = \begin{pmatrix} 8 & 6 & 9 & 5 \\ 6 & 9 & 5 & 10 \\ 5 & 8 & 4 & 9 \\ 10 & 6 & 11 & 4 \end{pmatrix}$$



$$I_1 = 0.4545\text{A and } I_2 = 0.1818\text{A}$$

